

BIOLOGICAL SYSTEMS ENGINEERING

For the undergraduate curriculum in Biological Systems Engineering leading to the degree Bachelor of Science. The Biological Systems Engineering Program is accredited by the Engineering Accreditation Commission of ABET, <https://www.abet.org> (<https://www.abet.org/>), under the commission's General Criteria and Program Criteria for Biological and Similarly Named Engineering Programs.

Biological Systems Engineers use their understanding of science and engineering to inspire designs and solve novel problems using natural systems as inspiration. Biological Systems Engineering integrates life sciences, biology, and chemistry with engineering to develop solutions for biological and ecological systems. Biological systems scale from microbes to ecosystems. They approach analysis and design from a systems perspective, appreciating the complexity of biological systems and developing solutions that accommodate and anticipate the adaptability of biological systems.

The Biological Systems Engineering program offers three focused degree options allowing students to craft their degree to match their career aspirations.

Ecological Engineering: Use biology, ecology, and engineering principles to design ecosystems for the mutual benefit of humans and nature. Enjoy a career that emphasizes the design, management, and restoration of ecosystems with roles in consulting firms, governmental agencies, and industry.

Food and Bioprocessing Engineering: Design and operate modern food processing systems, such as food pasteurization, sterilization, freezing, and hydration while improving the economic and environmental sustainability of biorenewable resource production systems. Enjoy a career improving the safety and quality of food by applying engineering, biology, and chemistry principles and emphasize the design and operation of industrial-scale bioconversion systems.

Open Option: Craft the degree to prepare for a professional degree program outside of engineering. Enjoy a career in academia, law, or industry that has a foundation in problem solving, systems approaches, biology, and chemistry.

Goal: To educate students to solve problems related to biorenewables production and processing, water quality, environmental impacts of the bioeconomy, food processing, and biosensors, and in so doing to prepare students for professional practice and post-graduate educational opportunities.

Student Learning Outcomes: Graduates of the Biological Systems Engineering curriculum should have at the time of graduation:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. An ability to communicate effectively with a range of audiences.
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives.
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Program Educational Objectives: Three to five years after graduation, our graduates will be using the knowledge, skills, and abilities from their biological systems engineering degree to improve the human condition through successful careers in a wide variety of fields. They will be effective leaders, collaborators, and innovators who address environmental, social, technical, and business challenges. They will be engaged in life-long learning and professional development through self-study, continuing education, or graduate/professional school.

Well-qualified juniors and seniors in biological systems engineering who are interested in graduate study may apply for concurrent enrollment in the Graduate College to simultaneously pursue a Bachelor of Science degree in biological systems engineering and a Master of Science degree in agricultural and biological engineering. Under concurrent enrollment, students are eligible for assistantships and simultaneously take undergraduate and graduate courses.

A concurrent Bachelor of Science and Master of Business Administration program is also offered by the department.

The department also offers a Bachelor of Science curriculum in agricultural engineering. See College of Engineering (<https://catalog.iastate.edu/collegeofengineering/>). Additionally, the department offers Bachelor of Science curricula in agricultural systems technology

and in industrial technology. See College of Agriculture and Life Sciences (<https://catalog.iastate.edu/collegeofagricultureandlifesciences/>).

The department also participates in interdepartmental majors in environmental science, sustainable agriculture, human computer interaction, and toxicology (see Index (<https://catalog.iastate.edu/azindex/>)).

Curriculum in Biological Systems Engineering

Administered by the Department of Agricultural and Biosystems Engineering.

Leading to the degree Bachelor of Science.

Total credits required:

129.0 cr Ecological Engineering Option

128.0 cr Food & Bioprocess Engineering Option

128.0cr Open Option.

Any transfer credit courses applied to the degree program require a grade of C or better (but will not be calculated into the ISU cumulative GPA, Basic Program GPA or Core GPA). See also Basic Program and Special Programs.

International Perspectives: 3 cr.¹

U.S. Cultures and Communities: 3 cr.¹

Communication Proficiency/Library requirement:

ENGL 1500	Critical Thinking and Communication (Must have a C or better in this course)	3
ENGL 2500	Written, Oral, Visual, and Electronic Composition (Must have a C or better in this course)	3
LIB 1600	Introduction to College Level Research	1
Communication Elective: One of the following (Must have a C or better in this course)		3
AGEDS 3110	Presentation and Sales Strategies for Agricultural Audiences	
ENGL 3090	Proposal and Report Writing	
ENGL 3140	Technical Communication	
MKT 4500	Advanced Professional Selling	
SPCM 2120	Fundamentals of Public Speaking	
SPCM 3120	Business and Professional Speaking	

Social Sciences and Humanities: 12 cr.^{1,2}

3 credits from international perspectives-university approved list	3
3 credits from U.S. Cultures and Communities approved list	3
6 credits from Social Sciences and Humanities courses-department approved list	6

Total Credits **12**

Basic Program: 24 cr.

A minimum GPA of 2.00 required for this set of courses (please note that transfer course grades will not be calculated into the Basic Program

GPA). See Requirement for Entry into Professional Program in College of Engineering Overview section. Within the Biological Systems Engineering Basic Program, students are required to complete CHEM 1670 and CHEM 1670L or the sequence of CHEM 1770, CHEM 1770L, and CHEM 1780. This is a departmental requirement within the College of Engineering Basic Program requirements. The CHEM 1780 course will show as completing the chemistry portion of the Basic Program and the credits will be applied towards a student's classification.

ABE 1600	Systematic Problem Solving and Computer Programming ³	3
CHEM 1670	General Chemistry for Engineering Students	4
or CHEM 1770 and CHEM 1780		
ENGL 1500	Critical Thinking and Communication (Must have a C or better in this course)	3
ENGR 1010	Engineering Orientation	R
LIB 1600	Introduction to College Level Research	1
MATH 1650	Calculus I	4
MATH 1660	Calculus II	4
PHYS 2310	Introduction to Classical Physics I	4
PHYS 2310L	Introduction to Classical Physics I Laboratory	1
Total Credits		24

Biological, Math and Physical Science: 14 cr.

BIOL 2120	Principles of Biology II	3
CHEM 1670L	Laboratory in General Chemistry for Engineering	1
or CHEM 1770L Laboratory in General Chemistry I		
MATH 2670	Elementary Differential Equations and Laplace Transforms	4
MICRO 3020	Biology of Microorganisms	3
STAT 3005	Engineering Statistics	3
Total Credits		14

Biological Systems Engineering Core: 42 cr.

(A minimum GPA of 2.00 required for this set of courses, including any transfer courses; please note that transfer course grades will not be calculated into the Core GPA).

ABE 2160	Fundamentals of Agricultural and Biosystems Engineering	3
ABE 2180	Project Management & Design in Agricultural and Biosystems Engineering	2
ABE 2730	CAD for Process Facilities and Land Use Planning	1
ABE 3160	Applied Numerical Methods for Agricultural and Biosystems Engineering	3
ABE 3630	Agri-Industrial Applications of Electric Power and Electronics	4

ABE 3780	Mechanics of Fluids	3
ABE 3800	Principles of Biological Systems Engineering	3
ABE 4040	Instrumentation for Agricultural and Biosystems Engineering	3
ABE 4150	Agricultural & Biosystems Engineering Design I	2
ABE 4160	Agricultural & Biosystems Engineering Design II	2
ABE 4800	Engineering Analysis of Biological Systems	3
CE 2710 & CE 2720 or CE 2740	Engineering Foundations of Statics and Applied Engineering Statics Engineering Statics	3
EM 3240	Mechanics of Materials	3
EM 3270 or ABE 3780L	Mechanics of Materials Laboratory Mechanics of Fluids Laboratory	1
IE 3050	Engineering Economic Analysis	3
ME 2310	Engineering Thermodynamics I	3

Total Credits 42

Other Remaining Courses: 11 cr.

ABE 1100	Experiencing Agricultural and Biosystems Engineering	1
ABE 1700	Engineering Graphics and Introductory Design	3
ABE 2010	Preparing for Workplace Seminar	1
ENGL 2500	Written, Oral, Visual, and Electronic Composition (Must have a C or better in this course)	3
Communication Elective: One of the following (Must have a C or better in this course)		3
AGEDS 3110	Presentation and Sales Strategies for Agricultural Audiences	
ENGL 3090	Proposal and Report Writing	
ENGL 3140	Technical Communication	
MKT 4500	Advanced Professional Selling	
SPCM 2120	Fundamentals of Public Speaking	
SPCM 3120	Business and Professional Speaking	

Total Credits 11

Complete remaining courses from one of the following options:

Ecological Engineering Option: 26 cr.

ABE 3340	Principles of Ecological Engineering	3
ABE 4310	Design and Evaluation of Soil and Water Conservation Systems	3
ABE 4340	Ecosystem Restoration Engineering	3
BIOL 3120	Ecology	4
CE 3720	Engineering Hydrology and Hydraulics	3
CHEM 2110	Quantitative and Environmental Analysis	2

CHEM 2110L	Quantitative and Environmental Analysis Laboratory	2
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Ecological Elective² 6

Total Credits 26

Food & Bioprocess Engineering Option: 25 cr.

ABE 4500	Advanced and Emerging Technologies in Biomanufacturing	3
ABE 4510	Food and Bioprocess Engineering	3
ABE 4690	Engineering for Grain Storage, Preservation, Handling, and Processing Systems	3
ME 4360	Heat Transfer	4
CHEM 2310	Elementary Organic Chemistry	3
CHEM 2310L	Laboratory in Elementary Organic Chemistry	1
FSHN 3110	Food Chemistry	3
FSHN 3110L	Food Chemistry Laboratory	1
MICRO 3000L	Introductory Microbiology Laboratory	1
Option Elective (select 3 cr from the following):		3
ABE 3250	Biorenewable Systems	
FSHN 4200	Food Microbiology	
FSHN 4710	Food Processing	
SCM 3010	Supply Chain Management	

Total Credits 25

Open Option: 25 cr.

ABE 4510	Food and Bioprocess Engineering	3
ME 4360	Heat Transfer	4
CHEM 3310	Organic Chemistry I	3
CHEM 3310L	Laboratory in Organic Chemistry I	1
CHEM 3320	Organic Chemistry II	3
CHEM 3320L	Laboratory in Organic Chemistry II	1
MICRO 3000L	Introductory Microbiology Laboratory	1
Sequence I, II & III Elective ²		9

Total Credits 25

Co-op/Internships (Optional)

- These university requirements will add to the minimum credits of the program unless the university-approved courses are also approved by the department to meet other course requirements within the degree program.
U.S. Cultures and Communities, International Perspectives and Social Science/Humanities courses may not be taken Pass/Not Pass.
- Choose from department approved list. (<http://www.abe.iastate.edu/undergraduate-students/biological-systems-engineering/bse-curricula/>)

3. See Basic Program for Professional Engineering Curricula for accepted substitutions for curriculum designated courses in the Basic Program.

TRANSFER CREDIT REQUIREMENTS

Students graduating with a degree in AE or BSE are required to have a minimum of 18 credits of 3000-level and 4000-level ABE courses taken at Iowa State University (excluding 4900, 4150, and 4160), and must complete the two-semester ABE Capstone sequence (ABE 4150 & 4160) at Iowa State University. The Department of Agricultural & Biosystems Engineering requires a grade of C or better for any transfer credit course that is applied to the degree program.

See also: A 4-year plan of study grid showing course template by semester for Biological Systems Engineering.

Biological Systems Engineering, B.S. - Ecological Engineering Option

First Year

Fall	Credits Spring	Credits
ENGR 1010	R ABE 1100	1
ABE 1700	3 ABE 1600	3
CHEM 1670	4 MATH 1660	4
CHEM 1670L	1 PHYS 2310	4
MATH 1650	4 PHYS 2310L	1
ENGL 1500	3 BIOL 2120	3
LIB 1600	1 CE 2710	1
	16	17

Second Year

Fall	Credits Spring	Credits
ABE 2160	3 ABE 2010	1
CE 2720	2 ABE 2180	2
CHEM 2110	2 MATH 2670	4
CHEM 2110L	2 STAT 3005	3
ENGL 2500	3 ABE 3780	3
BIOL 3120	4 ABE 3780L or EM 3270	1
	U.S. Communities & Cultures Elective	3
	16	17

Third Year

Fall	Credits Spring	Credits
ABE 2730	1 ABE 3630	4
ABE 3160	3 ABE 3800	3
ABE 3340	3 ABE 4310	3
CE 3720	3 EM 3240	3
ME 2310	3 IE 3050	3

Communication Elective	3	
	16	16

Fourth Year

Fall	Credits Spring	Credits
ABE 4150	2 ABE 4160	2
ABE 4040	3 Ecological Elective	3
ABE 4340	3 Ecological Elective	3
ABE 4800	3 Social Science or Humanities Elective	3
MICRO 3020	3 International Perspectives Elective	3
Social Science or Humanities Elective	3	
	17	14

Biological Systems Engineering, B.S. Food & Bioprocess Engineering Option

First Year

Fall	Credits Spring	Credits
ENGR 1010	R ABE 1100	1
ABE 1700	3 ABE 1600	3
MATH 1650	4 MATH 1660	4
CHEM 1670	4 PHYS 2310	4
CHEM 1670L	1 PHYS 2310L	1
ENGL 1500	3 ENGL 2500	3
LIB 1600	1 CE 2710	1
	16	17

Second Year

Fall	Credits Spring	Credits
ABE 2160	3 ABE 2010	1
CE 2720	2 ABE 2180	2
BIOL 2120	3 ABE 2730	1
CHEM 2310	3 MATH 2670	4
CHEM 2310L	1 MICRO 3020	3
U.S. Cultures and Communities Elective	3 MICRO 3000L	1
	ME 2310	3
	15	15

Third Year

Fall	Credits Spring	Credits
ABE 3160	3 ABE 3630	4
ABE 3780	3 ABE 3800	3
STAT 3005	3 ABE 4690	3

FSHN 3110	3	EM 3240	3	MICRO 3000L	1	IE 3050	3
FSHN 3110L	1	IE 3050	3	STAT 3005	3	Sequence I Elective	3
Communication Elective	3			Communication Elective	3		
	16		16		16		16

Fourth Year

Fall	Credits	Spring	Credits	Fall	Credits	Spring	Credits
ABE 4150	2	ABE 4160	2	ABE 4150	2	ABE 4160	2
ABE 4040	3	ABE 4510	3	ABE 4040	3	ABE 4510	3
ABE 4800	3	EM 3270 or ABE 3780L	1	ABE 4800	3	EM 3270 or ABE 3780L	1
ABE 4500	3	ME 4360	4	Sequence II Elective	3	ME 4360	4
Option Elective	3	Social Science or Humanities Elective	3	International Perspectives Elective	3	Sequence III Elective	3
International Perspectives Elective	3	Social Science or Humanities Elective	3	Social Science or Humanities Elective	3	Social Science or Humanities Elective	3
	17		16		17		16

Biological Systems Engineering, B.S. - Open Option

First Year

Fall	Credits	Spring	Credits
ENGR 1010	1	R ABE 1100	1
ABE 1700	3	ABE 1600	3
CHEM 1670	4	MATH 1660	4
CHEM 1670L	1	PHYS 2310	4
MATH 1650	1	PHYS 2310L	1
ENGL 1500	3	ENGL 2500	3
LIB 1600	1	CE 2710	1
	16		17

Second Year

Fall	Credits	Spring	Credits
ABE 2160	1	3 ABE 2010	1
CE 2720	2	ABE 2180	2
BIOL 2120	1	3 ABE 2730	1
CHEM 3310	3	CHEM 3320	3
CHEM 3310L	1	CHEM 3320L	1
U.S. Cultures and Communities Elective	4	3 MATH 2670	4
		ME 2310	3
	15		15

Third Year

Fall	Credits	Spring	Credits
ABE 3160	4	3 ABE 3630	4
ABE 3780	3	3 ABE 3800	3
MICRO 3020	3	3 EM 3240	3

Concurrent B.S./Master's Engineering Degree

The Department of Agricultural and Biosystems Engineering has concurrent Bachelor of Science /Master of Science (B.S./M.S.) program in agricultural and biosystems engineering designed especially for departmental seniors who wish to pursue advanced studies. Up to 6 credits of graduate course work can be used to satisfy your program requirements for both degrees

The concurrent B.S./M.S. program allows seniors to:

- Become eligible for a research assistantship during their senior year.
- Complete an M.S. degree (with thesis) within 18 months of B.S. graduation.

For more information about our concurrent undergraduate and graduate programs in Agricultural & Biosystems Engineering, visit:<https://www.abe.iastate.edu/graduate-students/abe-concurrent-degrees/>.

Concurrent B.S. in Biological Systems Engineering and MBA in the College of Business

Accelerate Your Career with an MBA

Today's successful engineers often need strong business management skills to complement their degree in engineering. These skills are especially important for engineers who are interested in working for technical companies, launching their own business ventures, or pursuing nontraditional jobs such as consulting. The MBA provides an opportunity for competitive advantage for advancement into management positions.

The Colleges of Business and Engineering offer an integrated, concurrent program leading to Bachelor of Science and Master of Business

Administration degrees. The program is designed to be completed in a minimum of five academic years or ten semesters. The actual time for completion may be longer depending upon semester course loads, internships, and course sequencing.

For more information about our concurrent undergraduate and graduate programs in Agricultural & Biosystems Engineering, visit:<https://www.abe.iastate.edu/graduate-students/abe-concurrent-degrees/>.

Graduate Study

The department offers Master of Science, Master of Engineering, and Doctor of Philosophy degrees with a major in agricultural and biosystems engineering. Within the agricultural and biosystems engineering major the student may specialize in advanced machinery engineering, animal production systems engineering, biological and process engineering, occupational safety engineering, or water and environmental stewardship engineering. For more information about our concurrent undergraduate and graduate programs in Agricultural & Biosystems Engineering, visit:<https://www.abe.iastate.edu/graduate-students/agricultural-engineering/>

For the Master of Science program, at least 30 credits of acceptable graduate work must be completed with a minimum of 22 credits of course work; corresponding numbers for the Master of Engineering program are 32 and 27. For the Doctor of Philosophy degree, at least 72 credits of acceptable graduate work must be completed with a minimum of 42 credits of course work. All Ph.D. students must complete a teaching/extension experience prior to graduation.

The department also offers both master of science and Doctor of Philosophy degrees in industrial and agricultural technology.

The department also participates in interdepartmental majors in environmental science, sustainable agriculture, human#computer interaction, and toxicology (see Index).